

Mirror Theory I

A Minimal Computational Theory of Recursive Observerhood

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Abstract

This paper develops a minimal computational theory of recursive observerhood. It asks what structural organisation must exist before the term observer can be used as a formal computational predicate. The proposed foundation consists of four primitive assumptions: distinguishability, dynamics, viability and bounded computational capacity. Symbols, representation, selfhood and observerhood are not introduced as primitives. They are treated as structures to be derived from persistence, compression, prediction and viability-constrained model selection.

The central result is an information-theoretic theorem. If future viability depends on compressed internal state, and if the reliability of the internal model carries additional information about future viability, then architectures maintaining compressed recursive self-models achieve strictly lower optimal predictive risk under log loss than architectures lacking those variables. A second theorem gives a threshold result. Under a selection rule favouring higher net viability and finite computational cost, recursive observerhood becomes the selected computational regime once the viability value of modelling self-model reliability exceeds its cost. The transition is expressed by a critical threshold and an order parameter for recursive self-modelling.

The framework is deliberately narrow. It gives a constructive account of recursive observerhood as a stable computational regime in bounded viability-constrained systems. It also proves a relative minimality result: within the derivational scheme used here, each primitive assumption is necessary for the stated theorem. The paper does not rely on semantic, psychological or metaphysical primitives. Its claims are conditional and falsifiable: if the specified information inequalities, compression assumptions or selection rule fail, the derivation fails at an identifiable point.

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1 Introduction

Many accounts of agency, learning and modelling begin after an observer has already been assumed. A system perceives, acts, predicts, remembers, updates or reports. In each case the explanatory role of the observer is already present. Mirror Theory asks an earlier computational question:

What is the smallest formal structure from which recursive observerhood can be derived?

The purpose of this paper is to state and prove a narrow structural claim. Recursive observerhood is treated as a computational regime of a bounded system that preserves viability by maintaining compressed models of its environment, compressed models of its own internal state, and compressed models of the reliability of those self-models. Nothing in the framework assumes that symbols, meanings, selves or observers exist as primitive entities. Each must either be derived from the computational setup or identified as a failure point.

The paper proceeds by construction. Section 2 introduces the mathematical notation. Section 3 states the primitive assumptions. Section 4 proves a relative minimality result for those assumptions. Section 5 derives persistence, tokenhood, compression, prediction, representation, self-modelling and recursive self-modelling. Section 6 defines predictive risk and net viability. Section 7 proves the information-theoretic lemmas needed for the main result. Section 8 proves the Recursive Observerhood Theorem. Section 9 proves the selection and threshold theorem. Section 11 proposes computational tests. Section 12 identifies the failure modes.

1.1 Contribution

The contribution is fourfold.

- (i) It defines recursive observerhood without taking observerhood as primitive.
- (ii) It proves an information-theoretic dominance theorem for compressed recursive self-models under explicit conditions.
- (iii) It proves a threshold theorem showing when recursive self-modelling becomes the selected stable computational regime under viability-constrained model selection.
- (iv) It proves a relative minimality theorem showing why none of the four primitive assumptions can simply be deleted without breaking the derivation.

The theory is not universal. It does not claim that all dynamical systems become observers. It identifies a class of systems for which recursive observerhood follows from stated assumptions. This restriction is essential: a theory that applies to every system explains too little. The target is a mathematically defined regime, not an all-purpose description of arbitrary complexity.

1.2 Methodological rule

The governing rule is the following.

No symbol, representation, self, identity or observer may be introduced as a primitive. Each must be derived from distinguishability, dynamics, viability and bounded computational capacity, or else the derivation fails at that point.

This rule is deliberately severe. It prevents the conclusion from being smuggled into the premises. In particular, when the paper uses terms such as representation or self-model, those terms refer only to formally defined structures with specified predictive and viability-preserving roles.

2 Mathematical Preliminaries

Let G denote a system embedded in an environment. Let $X_t \in \mathcal{X}$ denote the joint state at time t . Where useful, decompose the joint state as

$$X_t = (E_t, S_t), \quad (1)$$

where E_t denotes variables external to the system boundary being analysed and S_t denotes variables internal to that boundary. This decomposition does not assume a pre-given self. It marks the variables whose transitions are internal to the system under study.

A future viability variable is written

$$V_{t+1} = v(X_{t+1}), \quad (2)$$

where $v : \mathcal{X} \rightarrow \mathbb{R}$ is a viability function. When viability is treated as a set constraint, write $V \subseteq \mathcal{X}$. When it is treated as a scalar score, write $v(X_t) \in \mathbb{R}$.

For random variables A, B, C , conditional entropy is written $H(A | C)$, and conditional mutual information is written

$$I(A; B | C) = H(A | C) - H(A | B, C). \quad (3)$$

The paper uses log loss as the canonical predictive loss. If a predictive model outputs $q(F | Y)$ for a future variable F , then its expected log loss is

$$\mathbb{E}[-\log q(F | Y)]. \quad (4)$$

The optimal expected log-loss risk of variables Y is denoted

$$\mathcal{R}^*(Y) = \inf_q \mathbb{E}[-\log q(F | Y)]. \quad (5)$$

Definition 2.1 (Architecture). An architecture $M \in \mathcal{M}$ is a rule or mechanism for selecting, maintaining and updating internal variables used by a system for prediction or action. The variables available to architecture M are denoted Y_M .

Examples used throughout the paper are:

$$M_0 : Y_{M_0} = E_t, \quad (6)$$

$$M_1 : Y_{M_1} = (E_t, Z_t), \quad (7)$$

$$M_2 : Y_{M_2} = (E_t, Z_t, R_t), \quad (8)$$

where $Z_t = c(S_t)$ is a compressed internal variable and R_t is a reliability or error-profile variable concerning Z_t .

3 Primitive Axioms

The framework begins with four primitive assumptions. They are not presented as metaphysical foundations. They are the smallest current formal assumptions from which the derivation proceeds.

Axiom 3.1 (Distinguishability). There exists a state space $\mathcal{X}$ and a distinguishability relation

$$\delta : \mathcal{X} \times \mathcal{X} \rightarrow \{0, 1\}, \quad (9)$$

where $\delta(x, y) = 1$ means that states x and y are non-equivalent for the system under consideration.

Distinguishability supplies the minimal condition for structure. Without distinguishability, there are no differences for the dynamics to preserve, erase, compress or use.

Axiom 3.2 (Dynamics). There exists a transition rule Δ . In the deterministic case,

$$\Delta : \mathcal{X} \rightarrow \mathcal{X}. \quad (10)$$

In the input-dependent case,

$$\Delta : \mathcal{X} \times \mathcal{E} \rightarrow \mathcal{X}, \quad (11)$$

where $\mathcal{E}$ denotes environmental input or perturbation.

Dynamics supplies change. It makes persistence, recurrence, error and collapse possible.

Axiom 3.3 (Viability). There exists either a viability set $V \subseteq \mathcal{X}$ or a viability function

$$v : \mathcal{X} \rightarrow \mathbb{R}. \quad (12)$$

Viability specifies degrees of organizational preservation for the system under analysis.

Viability introduces a non-moral ordering over trajectories. A trajectory that preserves the relevant organization has higher viability than a trajectory that destroys it.

Axiom 3.4 (Bounded computational capacity). The system has finite computational capacity κ , representing finite memory, bandwidth, processing or representational capacity. If Z_t is an internal variable maintained by the system, then the admissible representations satisfy a capacity constraint such as

$$|Z_t| \leq \kappa. \quad (13)$$

Bounded capacity prevents exhaustive retention of arbitrary detail. It creates pressure for compression and model selection whenever recurrent structure is present.

4 Relative Minimality of the Primitive Axioms

The four primitive axioms are not claimed to be ontologically ultimate. The claim is more modest and more precise: relative to the derivational scheme of this paper, each primitive is necessary for deriving recursive observerhood in the stated sense. This section makes that claim explicit.

Definition 4.1 (Derivational scheme). The derivational scheme of the paper is the ordered construction

$$(\delta, \Delta, V, \kappa) \implies P \implies \tau \implies K \implies \Pi \implies \rho \implies Z \implies R \implies O, \quad (14)$$

where P denotes persistence, τ token formation, K compression, Π prediction, ρ predictive representation, Z compressed self-modelling, R recursive self-modelling and O recursive observerhood.

Definition 4.2 (Relative necessity). A primitive axiom is relatively necessary for the derivation if removing it makes at least one subsequent object in (14) undefined, vacuous or no longer derivable without introducing an equivalent replacement assumption.

Theorem 4.1 (Relative Minimality of the Primitive Basis). *Relative to Definition 4.1, each member of*

$$\{\delta, \Delta, V, \kappa\} \quad (15)$$

is necessary for deriving recursive observerhood as defined in Definition 5.12. Consequently, no proper subset of $\{\delta, \Delta, V, \kappa\}$ is sufficient for the derivation unless the missing primitive is reintroduced under another name.

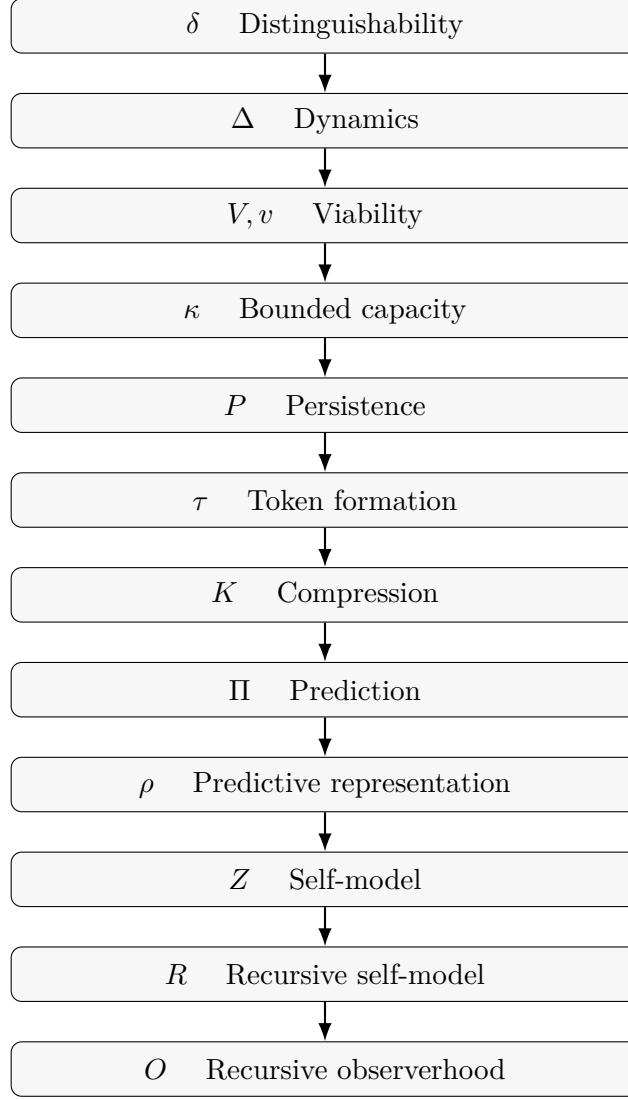


Figure 1: Primitive-to-observerhood dependency chain. Each arrow marks a derivational burden.

Proof. The proof proceeds by deletion.

Deletion of distinguishability. If δ is removed, there is no formal relation by which states are non-equivalent for the system. Persistence in Definition 5.1 is persistence of distinguishability across transitions. Without distinguishability, persistent distinctions are undefined. Since token formation is defined as an equivalence class of persistent distinctions, τ is also undefined. The chain therefore fails before compression, prediction or representation can be constructed.

Deletion of dynamics. If Δ is removed, there is no transition rule over which distinctions can persist or vanish. Persistence is therefore undefined, because $\delta(\Delta^n x, \Delta^n y)$ has no meaning. Prediction of future viability is also undefined because the future variable V_{t+1} depends on transition from X_t to X_{t+1} . Thus the derivation fails at persistence and again at prediction.

Deletion of viability. If V or v is removed, there is no future viability variable V_{t+1} , no viability-relevant information inequality, and no net viability functional $J(M)$. The terms predictively favoured and viably favoured can no longer be connected, because there is no criterion of organizational preservation for the system. The main theorem and the selection theorem both become undefined.

Deletion of bounded capacity. If κ is removed, compression is no longer forced by the formal setup. A system with unbounded capacity may retain arbitrary state detail without selecting

compressed internal variables. The specific object $Z_t = c(S_t)$ as a capacity-bounded compressed self-model is therefore not derivable as a response to computational constraint. Since recursive observerhood in Definition 5.12 is defined as a stable regime maintaining compressed self-models and models of their reliability, the derivation of recursive observerhood in the stated sense fails.

Each deletion breaks the construction unless an equivalent assumption is reintroduced. Hence each primitive is relatively necessary, and no proper subset suffices within this derivational scheme. $\square$

Remark 4.1 (Not an absolute impossibility theorem). Theorem 4.1 is a relative minimality theorem, not an absolute theorem about every conceivable formalism. A different theory might replace one primitive with a deeper structure from which it can be derived. The point is that, within the present framework, none of the four primitives is dispensable.

5 Derived Computational Structures

This section derives the structural objects used later in the theorems. The word “derive” is used in a conditional sense: each structure follows from the preceding structure plus any stated regularity condition. The paper does not claim that every system satisfies these conditions.

5.1 Persistence

A distinction is not yet a token. It becomes token-like only when it can be re-identified across transitions. Persistence is therefore the first derived structure.

Definition 5.1 (Persistent distinction). A distinction $\delta(x, y) = 1$ is persistent under Δ if its distinguishability is preserved across a non-trivial trajectory. In the deterministic case this may be written

$$\delta(\Delta^n x, \Delta^n y) = 1 \quad \text{for all } n \in \{0, \dots, T\} \quad (16)$$

for some $T > 0$. In a stochastic setting, persistence may be measured by

$$P(\delta) = \liminf_{n \rightarrow \infty} \mathbb{P}[\delta(\Delta^n x, \Delta^n y) = 1]. \quad (17)$$

Proposition 5.1 (Persistence is necessary for tokenhood). *Any later notion of tokenhood requires persistence under dynamics.*

Proof. A token must be re-identifiable across transformations. If no distinction is preserved across any non-trivial transition, then there is no invariant or equivalence class available for re-identification. Hence tokenhood cannot be defined without at least some persistent distinction. $\square$

5.2 Tokens as maintained invariants

Definition 5.2 (Token). Let $\mathcal{T}$ be a class of admissible transformations preserving persistence. A token is a persistent equivalence class

$$\tau = [\delta]_{\mathcal{T}}. \quad (18)$$

It is not a static object. It is a dynamically maintained invariant.

The point of this definition is modest. It does not say that a token is already a symbol. It says only that re-identifiable structure exists under dynamics.

5.3 Compression

Bounded capacity makes exhaustive memorisation unavailable. In recurrent environments, compressed variables can preserve viability-relevant regularities at lower cost than uncompressed state retention.

Definition 5.3 (Compression map). A compression map is a function

$$c : S \rightarrow Z, \quad (19)$$

where S is a space of internal states and Z is a lower-complexity internal variable satisfying $|Z_t| \leq \kappa$.

Definition 5.4 (Viability-relevant compression). A compressed variable $Z_t = c(S_t)$ preserves viability-relevant information in S_t relative to E_t if

$$I(V_{t+1}; Z_t \mid E_t) > 0. \quad (20)$$

It is sufficient for the viability-relevant component of S_t if

$$I(V_{t+1}; S_t \mid E_t, Z_t) = 0. \quad (21)$$

Equation (21) says that, once Z_t is known, the remaining details of S_t add no further information about future viability.

5.4 Prediction

Prediction enters because viability is temporally extended. A system that must preserve itself across transitions benefits from variables that reduce uncertainty about future viability, provided the reduced uncertainty can be used for action or internal regulation.

Definition 5.5 (Prediction variable). A variable Y is predictive of a future variable F when

$$I(F; Y) > 0. \quad (22)$$

It is conditionally predictive relative to C when

$$I(F; Y \mid C) > 0. \quad (23)$$

5.5 Predictive representation without an external interpreter

A persistent pattern is not automatically a representation. A whirlpool may persist without representing the river. A variable may also correlate with a future state without being used by the system. The framework therefore defines representation operationally: representation requires tracking, structure preservation and causal exploitation by the system's own predictive or viability-preserving dynamics.

Definition 5.6 (Tracking map). Let A be an internal substructure of system G , and let B be a target structure. A map $h : A \rightarrow B$ is a tracking map when changes in A covary with changes in B under the coupled dynamics and the map approximately preserves relevant transition structure:

$$h \circ \Delta_A \approx \Delta_B \circ h. \quad (24)$$

Definition 5.7 (Causal exploitation). An internal variable A is causally exploited by architecture M if replacing, randomising or ablating A , while holding the remaining model variables fixed, changes at least one of the following on a set of non-zero probability:

- (i) the predictive distribution $q_M(F \mid Y_M)$;
- (ii) the update rule for maintained model variables; or
- (iii) the policy, action or internal regulation rule available to the system.

The exploitation is viability-relevant when this change alters expected predictive risk or expected viability.

Definition 5.8 (Predictive representation). An internal substructure A is a predictive representation of target structure B for system G under architecture M if:

- (i) there exists a tracking map $h : A \rightarrow B$;
- (ii) h approximately preserves relevant transition structure as in (24); and
- (iii) A is causally exploited by M in a way that reduces predictive risk or increases expected viability for G .

Proposition 5.2 (Representation is stronger than passive correlation). *A variable that is merely correlated with a target structure but is not causally exploited by the system is not a predictive representation in the sense of Definition 5.8.*

Proof. By Definition 5.8, predictive representation requires causal exploitation. If ablating, replacing or randomising the variable does not change the predictive distribution, update rule, policy, action or internal regulation rule of the architecture on any set of non-zero probability, then the variable fails Definition 5.7. It therefore fails condition (iii) of Definition 5.8, even if it remains statistically correlated with the target. Hence passive correlation is insufficient for representation. $\square$

The causal exploitation condition prevents arbitrary homomorphisms or unused correlations from being misclassified as representations. A mapping becomes representational for the system only when it is tracked, structure-preserving and used by the system in a way that matters for prediction or viability.

5.6 Self-models

Definition 5.9 (Compressed self-model). A compressed self-model is a compressed internal variable

$$Z_t = c(S_t), \quad |Z_t| \leq \kappa, \quad (25)$$

that carries information about future viability not already contained in environmental variables:

$$I(V_{t+1}; Z_t \mid E_t) > 0. \quad (26)$$

This definition is purely computational. A self-model is not introduced as a psychological or semantic primitive. It is a viability-relevant compressed internal variable.

5.7 Recursive self-models

Definition 5.10 (Reliability variable). Let Z_t be a self-model. A reliability variable R_t is a variable encoding the reliability, calibration, confidence, error profile or performance condition of Z_t .

Definition 5.11 (Recursive self-model). A system maintains a recursive self-model when it maintains both Z_t and a reliability variable R_t concerning Z_t , and where the reliability variable carries additional information about future viability:

$$I(V_{t+1}; R_t \mid E_t, Z_t) > 0. \quad (27)$$

The word recursive is used in the limited sense that the system models a property of its own model. It need not contain an infinite regress. The recursion is finite and capacity-bounded.

Definition 5.12 (Recursive observerhood). Recursive observerhood is a stable regime in which a bounded viability-constrained system maintains compressed self-models and models of their reliability across persistence-relevant transitions.

6 Predictive Risk and Viability Selection

The paper uses two senses of “favoured”. The first is predictive. The second is viabilistic.

Definition 6.1 (Predictively favoured architecture). Architecture M_a is predictively favoured over architecture M_b with respect to future variable F when

$$\mathcal{R}^*(Y_{M_a}) < \mathcal{R}^*(Y_{M_b}). \quad (28)$$

Definition 6.2 (Net viability functional). For architecture M , define

$$J(M) = \mathbb{E}[v(X_{t+1}) \mid M] - \lambda \mathcal{C}(M), \quad (29)$$

where $\mathcal{C}(M)$ is computational cost and $\lambda \geq 0$ converts cost into viability-relevant penalty.

Definition 6.3 (Viably favoured architecture). Architecture M_a is viably favoured over architecture M_b when

$$J(M_a) > J(M_b). \quad (30)$$

The bridge from prediction to viability is a substantive condition. Improved prediction matters only if the prediction concerns a variable that the system can use to preserve viability.

Assumption 6.1 (Prediction-to-viability monotonicity). For the class of systems considered, reductions in optimal predictive risk concerning V_{t+1} produce non-negative expected viability gains, and produce strictly positive gains when the risk reduction concerns action-relevant or regulation-relevant viability information.

For the threshold theorem, this assumption is stated in a stronger local form.

Assumption 6.2 (Local viability gain bound). There exists $\beta > 0$ such that a reduction of predictive risk by η concerning V_{t+1} yields an expected viability gain of at least $\beta\eta$ over the local range being analysed.

The coefficient β is not a universal constant. It is a local lower-bound coefficient for the relevant system, environment and architecture class.

7 Information-Theoretic Lemmas

This section proves the information-theoretic engine of the paper.

Lemma 7.1 (Conditional information lowers conditional entropy). *Let A, B, C be random variables. If*

$$I(A; B \mid C) > 0, \quad (31)$$

then

$$H(A \mid B, C) < H(A \mid C). \quad (32)$$

Proof. By definition,

$$I(A; B | C) = H(A | C) - H(A | B, C). \quad (33)$$

If $I(A; B | C) > 0$, then the difference on the right-hand side is positive. Hence $H(A | C) > H(A | B, C)$, which is equivalent to (32). $\square$

Lemma 7.2 (Optimal log-loss risk equals conditional entropy). *Let F be a future variable and Y a set of model variables. Under log loss,*

$$\mathcal{R}^*(Y) = H(F | Y). \quad (34)$$

Proof. Let $p(F | Y)$ be the true conditional distribution and let $q(F | Y)$ be any proposed predictive distribution. The expected log loss decomposes as

$$\mathbb{E}[-\log q(F | Y)] = H(F | Y) + \mathbb{E}_Y[\text{KL}(p(F | Y) \| q(F | Y))]. \quad (35)$$

The Kullback-Leibler divergence is non-negative and equals zero if and only if $q(F | Y) = p(F | Y)$ almost surely. Hence the minimum over q is attained at the true conditional distribution, and the minimum value is $H(F | Y)$. $\square$

Lemma 7.3 (Internal-state information favours self-modelling). *Let V_{t+1} be future viability, E_t environmental state and S_t internal state. If*

$$I(V_{t+1}; S_t | E_t) > 0, \quad (36)$$

then predictors with access to the viability-relevant component of S_t achieve strictly lower optimal log-loss risk than predictors restricted to E_t alone.

Proof. Apply Lemma 7.1 with $A = V_{t+1}$, $B = S_t$ and $C = E_t$. The assumption gives

$$H(V_{t+1} | E_t, S_t) < H(V_{t+1} | E_t). \quad (37)$$

By Lemma 7.2, the two conditional entropies are the optimal log-loss risks of predictors using (E_t, S_t) and E_t respectively. Therefore internal-state information strictly lowers optimal predictive risk. $\square$

Lemma 7.4 (Compressed self-models suffice when viability information is preserved). *Let $Z_t = c(S_t)$ be a compressed internal variable with $|Z_t| \leq \kappa$. If*

$$I(V_{t+1}; Z_t | E_t) > 0, \quad (38)$$

then a predictor using (E_t, Z_t) has strictly lower optimal log-loss risk than a predictor using E_t alone. If Z_t is sufficient for the viability-relevant component of S_t , then the compressed variable loses no viability-relevant predictive advantage relative to that component.

Proof. Apply Lemma 7.1 with $A = V_{t+1}$, $B = Z_t$ and $C = E_t$. Then

$$H(V_{t+1} | E_t, Z_t) < H(V_{t+1} | E_t). \quad (39)$$

By Lemma 7.2, this is a strict reduction in optimal log-loss risk. If Z_t is sufficient in the sense of (21), then conditioning on S_t in addition to (E_t, Z_t) does not further reduce the uncertainty of V_{t+1} . Thus the compressed variable preserves the predictive advantage of the viability-relevant component of S_t . $\square$

Lemma 7.5 (Reliability information favours recursive self-modelling). *Let R_t denote the reliability or error-profile of the self-model Z_t . If*

$$I(V_{t+1}; R_t | E_t, Z_t) > 0, \quad (40)$$

then a predictor using (E_t, Z_t, R_t) has strictly lower optimal log-loss risk than a predictor using (E_t, Z_t) alone.

Proof. Apply Lemma 7.1 with $A = \mathbf{V}_{t+1}$, $B = R_t$ and $C = (E_t, Z_t)$. The assumption gives

$$H(\mathbf{V}_{t+1} \mid E_t, Z_t, R_t) < H(\mathbf{V}_{t+1} \mid E_t, Z_t). \quad (41)$$

By Lemma 7.2, this is a strict reduction in optimal predictive risk. Since R_t concerns the reliability or error-profile of Z_t , maintaining R_t is a finite model of a self-model. $\square$

8 Theorem 1: Recursive Observerhood

Theorem 8.1 (Recursive Observerhood). *Let G be a κ -bounded, viability-constrained dynamical system over distinguishable states. Let E_t denote environmental state, S_t internal state, $Z_t = c(S_t)$ a compressed self-model, R_t a reliability or error-profile variable concerning Z_t , and $\mathbf{V}_{t+1}$ future viability. Suppose that:*

- (i) $I(\mathbf{V}_{t+1}; Z_t \mid E_t) > 0$;
- (ii) $I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > 0$;
- (iii) *reductions in predictive risk concerning $\mathbf{V}_{t+1}$ are usable for viability preservation; and*
- (iv) *the cost of maintaining Z_t and R_t is finite and does not exceed the expected viability gain produced by their predictive advantage.*

Then architectures maintaining compressed recursive self-models are predictively and viably favoured over architectures lacking those variables. The resulting stable regime is recursive observerhood.

Proof. The proof has three steps.

Step 1: self-modelling. By assumption (i), Z_t contains information about future viability not contained in E_t alone:

$$I(\mathbf{V}_{t+1}; Z_t \mid E_t) > 0. \quad (42)$$

By Lemma 7.4,

$$H(\mathbf{V}_{t+1} \mid E_t, Z_t) < H(\mathbf{V}_{t+1} \mid E_t). \quad (43)$$

By Lemma 7.2, this is a strict reduction in optimal log-loss risk. Thus a compressed self-model is predictively favoured over an environment-only architecture whenever Z_t preserves viability-relevant internal information.

Step 2: recursive self-modelling. By assumption (ii), the reliability or error-profile of the self-model contains additional information about future viability:

$$I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > 0. \quad (44)$$

By Lemma 7.5,

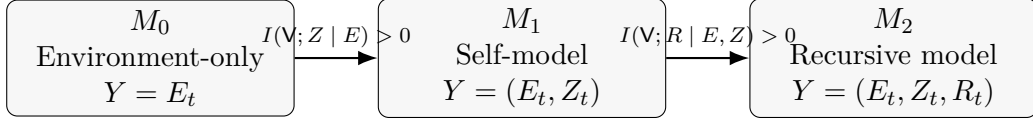
$$H(\mathbf{V}_{t+1} \mid E_t, Z_t, R_t) < H(\mathbf{V}_{t+1} \mid E_t, Z_t). \quad (45)$$

Again by Lemma 7.2, this is a strict reduction in optimal predictive risk. Since R_t is a variable concerning the performance or reliability of Z_t , maintaining R_t constitutes recursive self-modelling in the formal sense of Definition 5.11.

Step 3: viability. By assumption (iii), the predictive risk reductions in (43) and (45) are usable for preserving viability. By assumption (iv), the computational cost of maintaining Z_t and R_t does not exceed the expected viability gain produced by the predictive advantage. Therefore the recursively self-modelling architecture has higher net viability than the corresponding architecture lacking those variables:

$$J(M_{\text{recursive}}) > J(M_{\text{nonrecursive}}). \quad (46)$$

Hence compressed recursive self-models are both predictively and viably favoured. By Definition 5.12, a stable regime in which the system maintains such models across persistence-relevant transitions is recursive observerhood. $\square$



Selection rule: retain or reinforce architectures with higher $J(M)$.

Figure 2: Architecture ladder from environment-only prediction to recursive self-modelling.

Remark 8.1 (Scope of Theorem 1). The theorem is conditional. It does not assert that every bounded dynamical system becomes recursively self-modelling. It asserts that, when the information inequalities and viability-cost conditions hold, recursive self-modelling is not decorative: it is favoured by predictive and viability considerations.

9 Theorem 2: Selection and Threshold Transition

Theorem 8.1 proves that recursive self-modelling is favoured when the relevant information and cost conditions hold. This section proves a stronger selection result: once the recursive information advantage exceeds its cost, recursive observerhood becomes the selected computational regime under a specified selection rule.

9.1 Architecture classes

Define three architecture classes:

Architecture	Model variables	Computational regime
M_0	E_t	Environment-only prediction
M_1	(E_t, Z_t)	Compressed self-modelling
M_2	(E_t, Z_t, R_t)	Recursive self-modelling

Let $C_i = \mathcal{C}(M_i)$, and let

$$J_i = J(M_i) = \mathbb{E}[v(X_{t+1}) \mid M_i] - \lambda C_i. \quad (47)$$

Definition 9.1 (Selection operator). Given a finite architecture class $\mathcal{M}$, a hard viability selection operator is any rule

$$\mathcal{S}(\mathcal{M}) \in \arg \max_{M \in \mathcal{M}} J(M), \quad (48)$$

with a fixed tie-breaking convention.

Definition 9.2 (Recursive order parameter). For a selected architecture $\mathcal{S}(\mathcal{M})$, define

$$\Omega = \begin{cases} 0, & \text{if } \mathcal{S}(\mathcal{M}) \text{ is non-recursive,} \\ 1, & \text{if } \mathcal{S}(\mathcal{M}) \text{ is recursively self-modelling.} \end{cases} \quad (49)$$

9.2 Threshold conditions

The transition from M_0 to M_1 occurs when the net advantage of modelling compressed internal state exceeds its additional cost:

$$\beta I(\mathbf{V}_{t+1}; Z_t \mid E_t) > \lambda(C_1 - C_0). \quad (50)$$

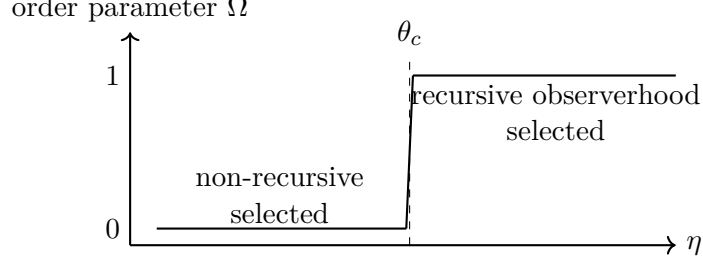


Figure 3: Hard-selection threshold transition into the recursive observerhood regime.

The transition from M_1 to M_2 occurs when the net advantage of modelling the reliability of the self-model exceeds its additional cost:

$$\beta I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > \lambda(C_2 - C_1). \quad (51)$$

The critical recursive threshold is therefore

$$\theta_c = \frac{\lambda(C_2 - C_1)}{\beta}. \quad (52)$$

The recursive architecture becomes net-favoured when

$$I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > \theta_c. \quad (53)$$

Theorem 9.1 (Selection Theorem for Recursive Observerhood). *Let G be a κ -bounded, viability-constrained dynamical system embedded in a recurrent environment. Suppose that:*

- (i) *persistent distinctions exist under the system dynamics;*
- (ii) *future viability depends on compressed internal state, so $I(\mathbf{V}_{t+1}; Z_t \mid E_t) > 0$;*
- (iii) *future viability also depends on the reliability or error-profile of the self-model:*

$$I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > 0; \quad (54)$$

- (iv) *reductions in predictive risk concerning future viability produce viability gains bounded below by $\beta > 0$;*
- (v) *computational cost is finite and penalised by $\lambda \geq 0$; and*
- (vi) *architectures with higher J are selected by $\mathcal{S}$ as in Definition 9.1.*

Then there exists a critical threshold

$$\theta_c = \frac{\lambda(C_2 - C_1)}{\beta} \quad (55)$$

such that, whenever

$$I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > \theta_c, \quad (56)$$

the recursive architecture M_2 is selected over the non-recursive self-modelling architecture M_1 . The selected regime has $\Omega = 1$ and is recursive observerhood.

Proof. Let M_1 be the best non-recursive self-modelling architecture using variables (E_t, Z_t) , and let M_2 be the corresponding recursive architecture using variables (E_t, Z_t, R_t) . By Lemma 7.5, the predictive risk reduction from adding R_t is

$$\Delta \mathcal{R}^* = H(\mathbf{V}_{t+1} \mid E_t, Z_t) - H(\mathbf{V}_{t+1} \mid E_t, Z_t, R_t) \quad (57)$$

$$= I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t), \quad (58)$$

where the second equality follows from the definition of conditional mutual information.

By assumption (iv), this predictive improvement produces an expected viability gain bounded below by

$$\beta I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t). \quad (59)$$

The additional computational cost of moving from M_1 to M_2 is

$$\lambda(C_2 - C_1). \quad (60)$$

Therefore the net viability advantage of M_2 over M_1 is bounded below by

$$J_2 - J_1 \geq \beta I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) - \lambda(C_2 - C_1). \quad (61)$$

If

$$I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > \frac{\lambda(C_2 - C_1)}{\beta}, \quad (62)$$

then $J_2 - J_1 > 0$. Hence M_2 has strictly greater net viability than M_1 . By assumption (vi), the selection operator selects the architecture with higher J . Therefore M_2 is selected over M_1 . Since M_2 is recursively self-modelling, the selected regime has $\Omega = 1$, which by Definition 5.12 is recursive observerhood. $\square$

Corollary 9.1 (Computational phase transition). *Under hard selection, the order parameter Ω changes from 0 to 1 as the control parameter*

$$\eta = I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) \quad (63)$$

crosses θ_c . This is a computational phase transition in selected regime.

Proof. For $\eta \leq \theta_c$, (61) does not force $J_2 > J_1$, so the recursive architecture is not selected by the theorem. For $\eta > \theta_c$, Theorem 9.1 gives $J_2 > J_1$, so M_2 is selected and $\Omega = 1$. The selected architecture changes at the threshold. $\square$

Remark 9.1 (Meaning of inevitability). The word inevitable is used only in this technical sense: within the specified class of systems, if the architecture is available and the information, cost and selection assumptions hold, recursive observerhood follows from the dominance of M_2 under J . No claim is made outside those assumptions.

10 Corollaries

Corollary 10.1 (Observerhood is not primitive). *If Theorem 8.1 holds, recursive observerhood need not be introduced as a primitive of the formal system.*

Proof. Theorem 8.1 derives recursive observerhood from distinguishability, dynamics, viability, bounded capacity and the stated information inequalities. Therefore observerhood is not required as a starting assumption within the formal framework. $\square$

Corollary 10.2 (Self-modelling is conditional on viability relevance). *If*

$$I(\mathbf{V}_{t+1}; Z_t \mid E_t) = 0, \quad (64)$$

then Z_t cannot be predictively favoured on the basis of future viability under log loss.

Proof. If the conditional mutual information is zero, then

$$H(\mathbf{V}_{t+1} \mid E_t, Z_t) = H(\mathbf{V}_{t+1} \mid E_t). \quad (65)$$

By Lemma 7.2, the optimal log-loss risks are equal. Hence no predictive advantage follows from adding Z_t . $\square$

Corollary 10.3 (Recursion is conditional on reliability relevance). *If*

$$I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) = 0, \quad (66)$$

then adding R_t cannot trigger the recursive threshold in Theorem 9.1.

Proof. The recursive threshold requires $I(\mathbf{V}_{t+1}; R_t \mid E_t, Z_t) > \theta_c$. If the left-hand side is zero and $\theta_c \geq 0$, the condition cannot be satisfied except in the degenerate case of zero threshold with strict inequality unavailable. $\square$

Corollary 10.4 (Ablation signature). *In an artificial system satisfying the assumptions, ablating R_t should reduce predictive performance or net viability only in environments where model reliability is viability-relevant and the recursive threshold is crossed.*

Proof. When R_t is viability-relevant and crosses the threshold, Theorem 9.1 selects M_2 over M_1 . Removing R_t forces the architecture back to M_1 , eliminating the predictive and viability advantage identified in (61). If R_t is not viability-relevant, Corollary 10.3 applies. $\square$

11 Computational Experiments

The theorems suggest direct computational tests. The experiments need not presuppose the conclusion. They require only agents with separable environmental variables, internal variables, compressed self-model variables and reliability variables.

Protocol 11.1 (Minimal ablation experiment). Construct a partially observable recurrent environment in which hidden conditions affect future viability. Compare three agents:

M_0 : environment-only prediction,

M_1 : environment plus compressed internal state,

M_2 : environment, compressed internal state and self-model reliability.

Evaluate optimal predictive risk, achieved predictive risk, computational cost and net viability.

Test	Expected result if theory holds	Failure interpretation
M_0 vs. M_1	M_1 outperforms M_0 when internal state is viability-relevant.	Z_t is not informative, not sufficient or not usable.
M_1 vs. M_2	M_2 outperforms M_1 when model reliability is viability-relevant.	R_t is irrelevant, too costly or poorly encoded.
Reliability relevance sweep	Selection shifts toward M_2 once $I(\mathbf{V}; R \mid E, Z) > \theta_c$.	Threshold model or selection assumption fails.
Random control	Compression and recursion provide little or no advantage.	Confirms dependence on recurrent structure if advantage disappears.

11.1 Implementation sketches

Several implementation substrates can test the formal structure.

Finite-state agents. Use a finite Markov decision process with hidden internal damage, calibration or energy variables. Let E_t be observed environmental state, S_t be internal resource state, Z_t be a compressed estimate of S_t , and R_t be the error estimate of Z_t . Vary the degree to which errors in Z_t affect future viability.

Graph rewriting systems. Represent distinctions as labelled graph differences. Dynamics rewrite the graph. Persistent motifs become tokens. Compression maps motifs into lower-dimensional signatures. Self-models are subgraphs encoding internal state. Recursive self-models are subgraphs encoding the reliability of those encodings.

Learning agents. Train three agents with matched capacity budgets: one without internal-state variables, one with compressed internal-state variables, and one with explicit reliability tracking. Evaluate whether the recursive architecture is selected only when reliability becomes viability-relevant and cost-effective.

11.2 Measurement criteria

The empirical signature of the theory is not mere performance improvement. The expected signature is structured:

- (i) self-modelling should help only when internal state contains viability-relevant information;
- (ii) recursive self-modelling should help only when model reliability contains additional viability-relevant information;
- (iii) increasing the relevance of reliability should produce a threshold-like shift in selected architecture;
- (iv) increasing computational cost should move the threshold upward as predicted by $\theta_c = \lambda(C_2 - C_1)/\beta$.

12 Failure Analysis

The framework is strongest where semantic-sounding language is reduced to dynamics, information and viability. It is weakest where the derivations require additional assumptions. The main failure points are as follows.

Viability may be a substantial primitive. Distinguishability and dynamics alone do not contain a notion of better or worse for a system. Viability may need to be justified by a broader theory of self-maintaining dynamical organization. In this paper it is treated as a primitive constraint.

Compression depends on recurrent structure. In maximally random environments, compression has little value. The derivation applies to systems embedded in environments with exploitable regularity.

Representation may remain edge-case sensitive. A mapping may be predictive without satisfying every intuitive use of the word representation. The paper therefore defines representation operationally: tracking plus structure preservation plus predictive or viability contribution.

Bounded capacity does important work. The capacity bound κ is not a cosmetic assumption. It drives compression and model selection. If capacity is unbounded, the pressure toward compressed modelling weakens. This is one reason the minimality theorem is stated relatively: a different framework might derive compression from another resource constraint, but this paper requires an explicit capacity bound.

Prediction-to-viability monotonicity can fail. A system may predict future viability but be unable to use the prediction. In that case the predictive result remains true, but the viability result does not follow.

The selection rule is essential. The threshold theorem requires a rule that retains, reinforces or selects architectures with higher J . Without such a rule, higher net viability does not force architectural dominance.

The phase transition may be smoothed in implementation. The theorem uses a hard selection operator. Real systems may approximate selection through gradual learning, reinforcement or population dynamics. In those cases Ω may change smoothly rather than discontinuously, but the threshold inequality still identifies the point at which recursive modelling becomes net-favoured.

13 Conclusion

This paper has developed Mirror Theory as a minimal computational theory of recursive observerhood. Starting from distinguishability, dynamics, viability and bounded computational capacity, it derived persistence, token formation, compression, prediction, predictive representation, compressed self-modelling and recursive self-modelling. It also proved that, relative to the derivational scheme used here, each of the four primitives is necessary for the derivation.

The first main theorem proved that, where future viability depends on compressed internal state and on the reliability of the self-model, recursive self-modelling strictly improves optimal predictive risk under log loss and is viably favoured when the viability gain exceeds computational cost. The second main theorem proved a threshold result: under a viability selection rule, recursive observerhood becomes the selected regime once the information value of modelling self-model reliability exceeds its computational cost.

The theory is conditional and constructive. If internal state is not viability-relevant, self-modelling is not derived. If self-model reliability is not viability-relevant, recursion is not derived. If computational cost exceeds predictive benefit, recursive observerhood is not selected. The value of the framework lies in making these dependencies explicit.

A Information-Theoretic Identities

For random variables A, B, C :

$$I(A; B \mid C) = H(A \mid C) - H(A \mid B, C), \quad (67)$$

$$I(A; B \mid C) = H(B \mid C) - H(B \mid A, C), \quad (68)$$

$$I(A; B \mid C) \geq 0. \quad (69)$$

For log loss:

$$\mathbb{E}[-\log q(F \mid Y)] = H(F \mid Y) + \mathbb{E}_Y[\text{KL}(p(F \mid Y) \parallel q(F \mid Y))]. \quad (70)$$

Since $\text{KL}(p \parallel q) \geq 0$, the optimal predictive distribution under log loss is the true conditional distribution $p(F \mid Y)$, and the optimal value is $H(F \mid Y)$.

B Notation

Symbol	Interpretation
$\mathcal{X}$	State space
X_t	Joint state at time t
E_t	Environmental component of state
S_t	Internal component of state
δ	Distinguishability relation
Δ	State transition rule
V	Viability set
v	Viability function
V_{t+1}	Future viability variable $v(X_{t+1})$
κ	Bounded computational capacity
P	Persistence
τ	Token as persistent equivalence class
c	Compression map
Z_t	Compressed self-model
R_t	Reliability or error-profile of Z_t
$I(\cdot; \cdot \cdot)$	Conditional mutual information
$H(\cdot \cdot)$	Conditional entropy
$\mathcal{R}^*$	Optimal expected predictive risk
$\mathcal{C}(M)$	Computational cost of architecture M
$J(M)$	Net viability functional
β	Local lower-bound coefficient from risk reduction to viability gain
λ	Cost penalty coefficient
θ_c	Critical recursive threshold
Ω	Order parameter for recursive observerhood regime

C Selected References

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